Lab 03

1. The provided code uses *static* methods to calculate a student's average grade and determine their letter grade (***procedural programming***). Your task is to refactor this implementation into an ***object-oriented*** approach.

```
static double calculateAverage(int[] grades) {
    int sum = 0;
    for (int grade : grades) {
      sum += grade;
    }
    return (double) sum / grades.length;
}

static char determineGrade(double average) {
  if (average >= 90) return 'A';
    else if (average >= 80) return 'B';
    else if (average >= 70) return 'C';
    else if (average >= 60) return 'D';
    else return 'F';
}
```

Instructions:

a) Create a `**Student**` class that:
- Stores the student's name and grades
- Has constructor `Student(String name, int[] grades)`
- Has a method `` `calculateAverage()` `` to compute the average grade.
- Has a method `` `determineGrade()` `` to determine the letter grade.
- Has a method `` `getName()` `` to retrieve the student's name.

b) In the `Main` class, create 3 `**Student**` objects and use it to:
- Calculate and print the student's average grade.
- Determine and print the student's letter grade.

2. Now that we have converted the procedural approach to an object-oriented design, we will explore when/how to use static members correctly.

Extend the `Student` class by:

a) Adding a static variable `**TopStudentGrad**` to store the highest averageGrade among all students.
b) Adding a static variable `**TopStudentName**` to store the name of the student with the highest grade.
c) Implementing a static method `**getTopStudentName**()` to return the name of the student with the highest grade.
d) Updating the constructor to check if the new student has a higher grade than `**TopStudentGrade**`, and update `**TopStudentName**` accordingly.

e) Try the following statements in the method **getTopStudentName**():
   a. `System.out.println(name);` // Does it work, comment!
   b. `System.out.println(getName());` // Does it work, comment!
   c. `System.out.println(topStudentName);` // Does it work, comment!

f) Try the following statements in the method **determineGrade**():
   a. `System.out.println(name);` // Does it work, comment!
   b. `System.out.println(getName());` // Does it work, comment!
   c. `System.out.println(TopStudentName);` // Does it work, comment!

Try the following code statements in the main method that **uses** the student class

a. `Student s = new Student("Alice", new int[]{90, 95, 85});`
b. `System.out.println(s.getTopStudentName());`
   // Does it work, comment!
c. `System.out.println(Student.getTopStudentName());`
   // Does it work, comment!
d. `System.out.println(s.getName());`
   // Does it work, comment!
e. `System.out.println(Student.getName());`
   // Does it work, comment!